

Practice Quiz: Cognition (Implementation)

Name: __ Date: __

Part A: Multiple Choice

1. The T-Maze has how many hidden states? A) 2 B) 4 C) 8 — (4 locations $\times$ 2 reward conditions) D) 16
2. The T-Maze uses 3 observation modalities because: A) Three is a magic number B) The agent needs separate channels for location, cue, and reward information C) Python requires 3 arrays D) There are 3 states
3. The expected behavior of an Active Inference agent in the T-Maze is: A) Always go left B) First visit the cue (information-seeking), then go to the rewarded arm (goal-seeking) C) Never visit the cue D) Stay in the center forever
4. The agent visits the cue because: A) It's programmed to B) The cue has high epistemic value — visiting it reduces uncertainty about the reward location C) The cue has high reward D) There's no other option
5. If cue reliability is 0.5 (random), the agent: A) Still visits the cue out of habit B) Has less incentive to visit the cue since it provides no information C) Always visits the cue D) Crashes
6. The `C_reward` vector = [3, -3, 0] means: A) Reward has value 3, loss has penalty -3, neutral is neither B) There are 3 states C) The agent prefers 3 rewards D) The discount factor is 3
7. In the B matrix, the reward condition dimension doesn't change because: A) The code is incomplete B) The reward is permanently in one location — the agent's actions don't move the reward C) B only encodes location D) NumPy doesn't support 4D arrays

Part B: Short Answer

1. Explain why the T-Maze demonstrates the exploration-exploitation balance. What drives the agent to explore (visit cue) vs. exploit (go to reward arm)?

2. Design a modification to the T-Maze where there are 3 arms instead of 2, with cues of different reliability for each arm. How would A, B, C, D change?
3. What happens to the agent's behavior if you set $C_reward = [0, 0, 0]$? Why? What is the agent optimizing for in the absence of preferences?